

## Multiple Choice

1. **Answer:** D

*Options:* A) 15, 17; B) 15, 19; C) 17, 19; D) 17, 21

*Note:* The correct answer is 17, 21 because the pattern increases by 4 with each successive term, and continuing this pattern from 13 results in 17 and then 21.

2. **Answer:** B

*Options:* A)  $1850x$ ; B)  $2000 - 150x$ ; C)  $150x$ ; D)  $2000 + 150x$

*Note:* The expression  $2000 - 150x$  correctly represents the total amount of money the Sojourn family has left after spending 150 each day for  $x$  days, starting with an initial amount of 2000.

3. **Answer:** B

*Options:* A)  $3k+1$ ; B)  $5k$ ; C)  $3k$ ; D) 1

*Note:* The correct answer is  $5k$  because simplifying the expression involves combining like terms, where the fractions are converted to a common denominator and then combined, leading to a final result of  $5k$ .

4. **Answer:** B

*Options:* A) add 6 to both sides; B) divide both sides by 6; C) multiply both sides by 6; D) subtract 6 from both sides

*Note:* To isolate the variable  $a$  in the equation  $6a = 72$ , you must perform the inverse operation of multiplication by dividing both sides of the equation by 6.

5. **Answer:** D

*Options:* A) 11.9 in.; B) 11 in.; C) 45.0 in.; D) 12 in.

*Note:* By rounding each rainfall measurement to the nearest whole number (8 inches, 4 inches, and 0 inches), the estimated total rainfall becomes 12 inches.

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* The expression  $9(9m + 3t)$  simplifies to  $81m + 27t$ , not  $18m + 12t$ , because when distributing 9, each term inside the parentheses must be multiplied by 9.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* The statement is true because the cost per stamp is  $8.28 \div 18 = 0.46$ , and therefore, for 12 stamps, the total cost is  $12 \times 0.46 = 5.52$ .

8. **Answer:** False

*Options:* A) True; B) False

*Note:* The ratio 12 over 27 simplifies to 4 over 9, while 8 over 18 simplifies to 4 over 9 as well, so they are equivalent ratios, making the statement true, not false.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* The value of 'a' is -4 because for two lines to be perpendicular, the product of their slopes must equal -1, and the slope of one line is the negative reciprocal of the other, which is satisfied by this value.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* The total cost of 934 puzzles at 6 each is actually 5,604, calculated by multiplying 934 by 6, which makes the statement false.

## Long Form Response

11. **Answer:** Let  $s$  be the side length of the square. Then the dimensions of each rectangle are  $s \times \frac{s}{5}$ . The perimeter of one of the rectangles is  $s + \frac{s}{5} + s + \frac{s}{5} = \frac{12}{5}s$ . Setting  $\frac{12}{5}s = 36$  inches, we find  $s = 15$  inches. The perimeter of the square is  $4s = 4(15 \text{ in.}) = 60$  inches. *Note:* correct because it accurately derives the side length of the square from the given perimeter of the rectangles, leading to the correct calculation of the square's perimeter.

12. **Answer:** To find the 8 th number, we start by calculating the total of the first 7 numbers. Since their mean is 15, the total is 7 times 15, which equals 105. When we add the 8 th number, the new mean becomes 12 for all 8 numbers. Therefore, the total for these 8 numbers is 8 times 12, which equals 96. To find the 8 th number, we subtract the total of the first 7 numbers from the total of all 8 numbers:  $96 - 105 = -9$ . Thus, the 8 th number is -9.

*Note:* correct because it accurately calculates the total of the first 7 numbers using the mean, determines the total for all 8 numbers based on the new mean, and then finds the 8 th number by subtracting the total of the first 7 from the total of all 8, resulting in -9.

*End of Answer Key*